



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

 Website: www.ktunotes.in

ie $s(s+p_1)(s+p_2) + k(s+z_1)(s+z_2)$

Roots of CE is a function of open loop gain k . When the k is varied from 0 to ∞ roots will take different values.

When $k=0$

roots = OL poles ie $(s=0, -p_1, -p_2)$

When $k=\infty$

roots = OL zeros

ie ① $\div k \frac{s(s+p_1)(s+p_2) + (s+z_1)(s+z_2)}{k}$

put $k=\infty$

$$(s+z_1)(s+z_2)=0$$

$$\text{ie } s = -z_1, -z_2$$

The path taken by roots of CE as OL gain k is varied from 0 to ∞ are called root loci.

(Roots of CE can be varied by any other system parameter other than k).

$$|G(s)H(s)| = 1$$

2) Angle criterion

$$\angle G(s)H(s) = \pm 180^\circ (2q + 1)$$

magnitude criterion states

$S = S_a$ will be a point on root locus if for that value of S

$$|G(s)H(s)| = 1$$

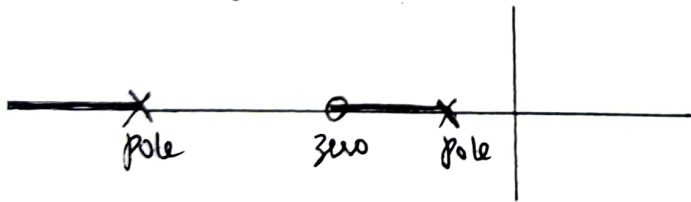
The angle criterion states

$S = S_a$ will be a point on root locus if for that value of S , $\angle G(s)H(s)$ is equal to an odd multiple of 180° .

4) No. of branches
 No. of branches of RL is
 if $P > Z$ & $N = Z$ if $Z > P$

5) existence on real axis

The existence of a section of real axis is no if the sum of OL poles & zero the right of this section is



6) Break away point

on the RL b/w 2
 the roots move towards each other as K is increased till they are coincident. Any further increase in K breaks the root locus into 2 parts. The break away point can be determined by

$$\frac{(2k+1)180}{P-Z} \quad \text{where } k = 0, 1, 2, \dots$$

8) Intersection of asymptote
real axis (centroid).

$$\sigma = \frac{\sum \text{poles} - \sum \text{zeros}}{P-Z}$$

9) Intersection point with
axis

The pt at which
root locus crosses the
is determined by RH cr
By applying RH criteria

10) Angle of departure from

The angle of de
of the RL from a complex
given by

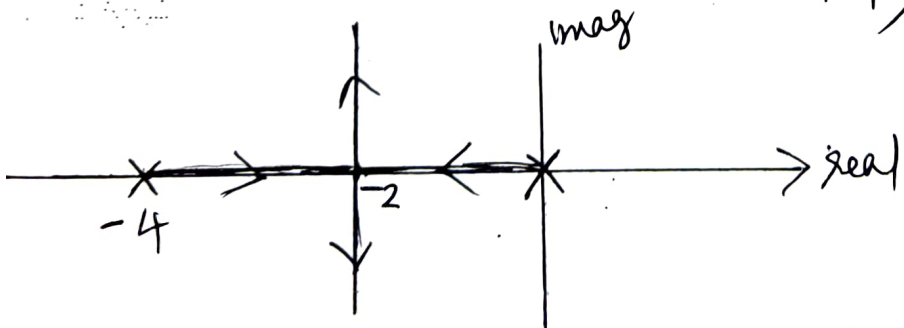
$$\phi_d = 180 - (\phi_p - \phi_z)$$

is the sum of all angles
subtended by remaining

ϕ_p - sum of all angles ^{by remaining} $\angle$ by poles.
 angle of arrival is tang ^{by} to the RL at the complex.

Effect of addition of pole on root locus

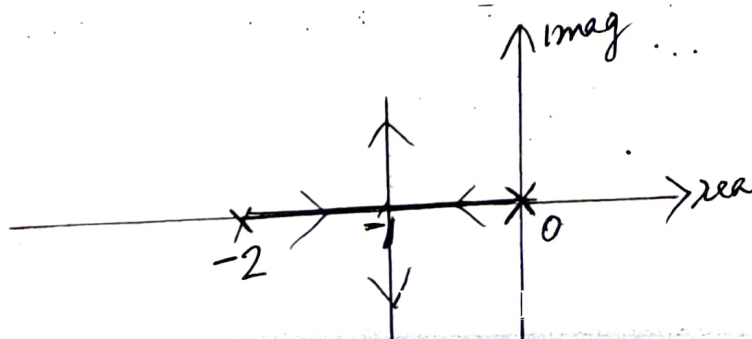
consider $G(s)H(s) = \frac{k}{s(s+4)}$



Now add one pole at $s = -5$
 $G(s)H(s) = \frac{k}{s(s+4)(s+5)}$

one equal number of poles
 left half, the two branches
 root locus moves to the right
 for some value of K . $\therefore$ s) $\therefore$
 be stable for this value
 this value of K s) m becom
 unstable. So stability of
 gets restricted. For further
 of poles to left half the
 breaking point moves toward
 $\therefore$ we can say that by
 of poles to left half,
 RL shifted towards right
 side of stability of s) m

Now consider OLTF $G(s)$



By addition of zeros towards the RL shifts towards the relative stability increases

ie by addition of poles, ϕ towards imag axis & ζ/ω_n decreases, while by adding zeros towards left half away from imaginary axis ζ/ω_n stability increases

root locus gives S_d .

- 3) Determine the open loop gain uncompensated s/m at
let this gain be K OL g
at $S = S_d$ on RL is given
 $K = \frac{\text{Prdt of vector lengths from } S_d}{\text{Prdt of vector lengths from } S_d}$

- 4) calculate β of compensator
 $K_{vu} = \text{vel error constant of}$

$K_{vd} = \text{desired velocity error constant}$

$$K_{vu} = \lim_{s \rightarrow 0} s G(s)$$

'A' be factor by which vel
const of s/m has to be

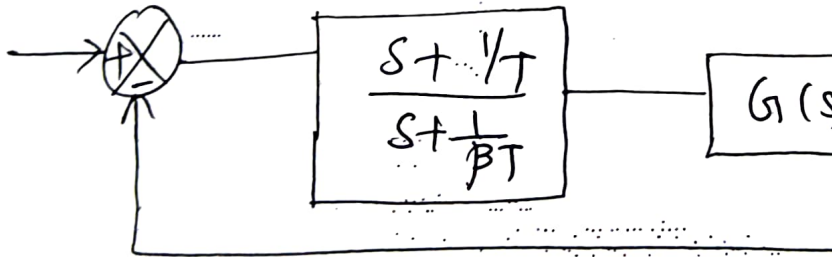
$$A = \frac{K_{vd}}{K_{vu}}$$

$\beta = 10$ to 20% greater than
 $\beta = (1.1 \text{ to } 1.2) A$

TF of lag comp $G_c(s) = \frac{s + \dots}{s + \dots}$

$$= \beta \frac{(1 + sT)}{(1 + s\beta T)}$$

6)



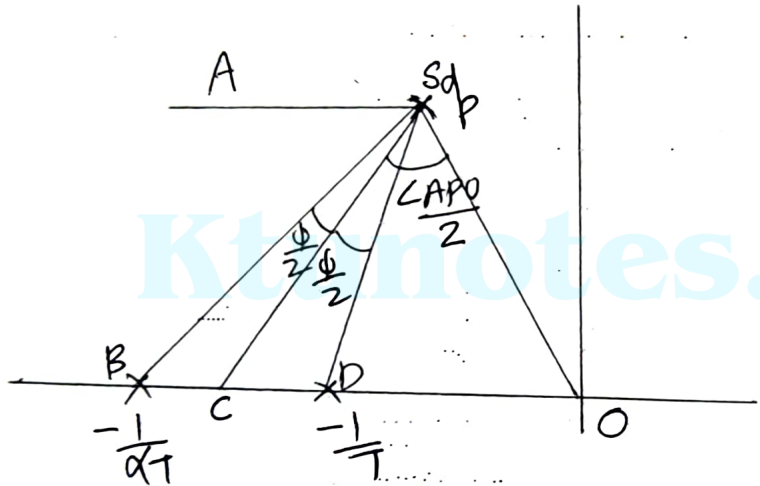
lag compensated s/

$$G_0(s) = G_c(s) G(s) \text{ is}$$

of compensated

7) Check whether s/m satisfies steady state error req. otherwise repeat design modifying the location poles & zeros of comp

8 S_d on S plane.



3) Determine ϕ angle to be
by lead n/w to make the
as point on RL

$$\phi = \left(\text{sum of angles contributed by poles of uncomp sim} \right) - \left(\text{sum of angles contributed by zeros of uncomp} \right)$$

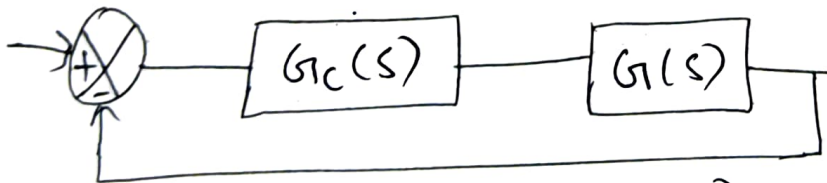
n is an odd integer

Draw vectors from all OL
zeros to point P. measure
contributed by vectors

to $\pi/2$. B & D on real axis
 the location of pole of Com
 and D zero $-\frac{1}{T}$

From values of ρ & B & D
 T & α

$$5) G_c(s) = \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}} = \frac{\alpha(1 + sT)}{1 + s\alpha T}$$



6) OLTF of Comp s/m $G_0(s) = G_c(s)$
 Determine K

$K = \text{Prdt of vector lengths from}$

17

zeros

7) Check whether s/m sat
 error requirement.

1) determine S_d

$$S_d = -\delta \omega_n \pm j \omega_n \sqrt{1 - \delta^2}$$

2)

3)

4) same as before

$$5) \quad G_2(s) = \frac{s + \frac{1}{T_2}}{s + \frac{1}{\alpha T_2}}$$

6) determine OL gain K

$K = \frac{\text{Prdt of vec length from a}}{\text{zeros to}}$

OLTF of lead comp s/m

$$G_{O2}(s) = G_2(s) G_1(s)$$

7) vel error const of lead

$$K_{V2} = \lim_{s \rightarrow 0} s G_{O2}(s)$$

if K_{V2} satisfies requirement
only lead section sufficient

$$Z_{C1} = -\frac{1}{T_1}$$

$$\therefore T_1 = -\frac{1}{Z_{C1}}$$

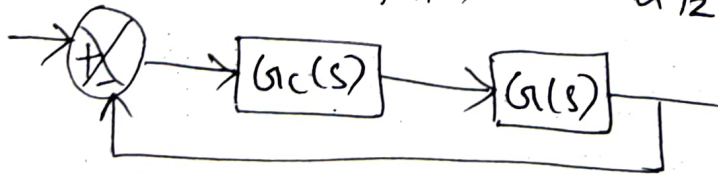
Pole of lag section $P_{C1} =$

TF of lag section $G_1(s) =$

10) TF of lag lead comp

$$G_c(s) = G_1(s) G_2(s)$$

$$= \frac{(s + \frac{1}{T_1})(s + \frac{1}{T_2})}{(s + \frac{1}{\beta T_1})(s + \frac{1}{\alpha T_2})}$$

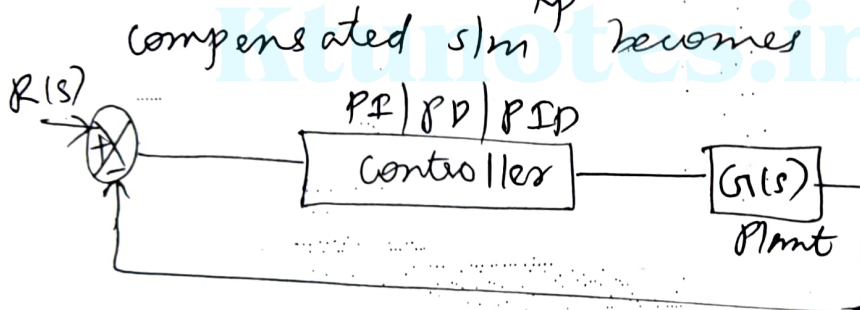


OLTF of lag lead comp $s/m = G_0(s) = G$

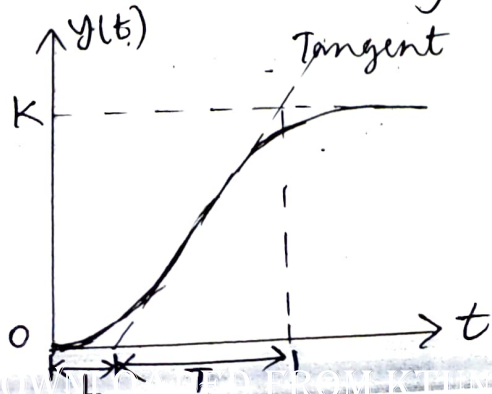
11) check kv of comp s

$$T_i = \text{reset time} = \frac{k_p}{k_i}$$

$$T_d = \text{rate time or derivative} = \frac{k_d}{k_p}$$



First method of ZN Tuning for PI
 Dynamic model of plant not known
 Applied to plants with step response
 of form shown in fig



Typical
response
curve

response is typical of a 1st order system with transportation delay. The flow from tank along a pipe is an 'S' shaped curve is characterized by two constants, the delay time 'L' and time constant 'T' from drawing tangent & noting intersections with the time axis. The steady state value 'K'.

TF of plant may be approximated by a first order system as

$$\frac{C(s)}{R(s)} = G(s) = \frac{K e^{-LS}}{1 + ST}$$

The values of K_p , T_i & T_d suggested by Ziegler-Nichols are given in table

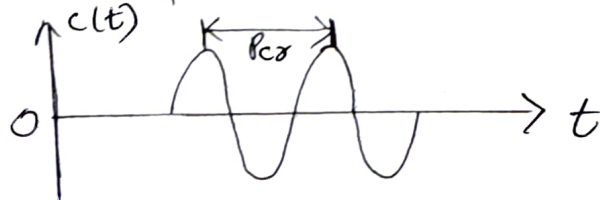
Type of controller	K_p	$T_i = K_p / k_i$	$T_d = k_d / k_i$
P	T/L	∞	0
PI	$0.9 T/L$	$\frac{L}{0.3}$	0
PID	$1.2 T/L$	$2L$	$0.5L$

Second method (oscillatory system)

Dynamic model of the plant is
In this method first set $T_i = 0$
 $T_d = 0$ i.e. controller has
proportional gain k_p . set value
such that o/p exhibits oscillations

Steps

- 1) Reduce integrator & derivative to zero.
- 2) Increase k_p from 0 to critical
 $k_p = K_{cr}$ at which sustained
oscillations occur.
- 3) The critical gain k_{cr} & corresponding
period P_{cr} are experimentally
determined as shown in



$$= 0.6 K_{cr} \left(1 + \frac{1}{0.5 P_{cr} S} + 0.12 \right)$$

$$G_c(s) = 0.075 K_{cr} P_{cr} \left(s + \frac{4}{P_{cr}} \right)$$

PID controller has pole at $s = 0$ & a double zero at $s =$

if mathematical model known can use RL to find K_{cr} & frequency of oscillation ω_{cr}

$$\frac{2\pi}{\omega_{cr}} = P_{cr} \cdot \text{These values}$$

obtained by determining n of RL with imaginary

$$(s+1)(s+2)(s+3)$$

$$CE \Rightarrow 1 + kp G(s) = 0$$

$$1 + \frac{6kp}{s^3 + 6s^2 + 11s + 6} = 0$$

$$s^3 + 6s^2 + 11s + 6 + 6kp = 0$$

$$s^3 + 6s^2 + 11s + 6(1+kp) = 0$$

with array

$$s^3 \quad 1 \quad 11$$

$$s^2 \quad 6 \quad 6(kp+1)$$

$$s^1 \quad 11-(kp+1)$$

$$s^0 \quad 6(kp+1)$$

$$6(kp+1) > 0 \Rightarrow 6kp > -6$$

$$11-(kp+1) > 0 \Rightarrow 10 - kp > 0$$

$$\text{range } 0 < kp \leq 10$$

$$\text{critical gain } k_{cr} = 10$$

$$\text{auxiliary polynomial } 6s^2 + 6(1+kp)$$

$$(s+1)(s+2)(s+3)$$

$$TF = \frac{k_p G(s)}{1 + k_p G(s)} = \frac{30}{s^3 + 6s^2 + 11s}$$

PI control

$$k_p = 0.45 k_{cr} = 4.5$$

$$T_i = \frac{P_{cr}}{1.2} = \frac{1.895}{1.2} = 1.579$$

$$G' = k_p \left(1 + \frac{1}{T_i s}\right) G(s) = 4.5 \frac{(s+1)(s+2)(s+3)}{s^3 + 6s^2 + 11s}$$

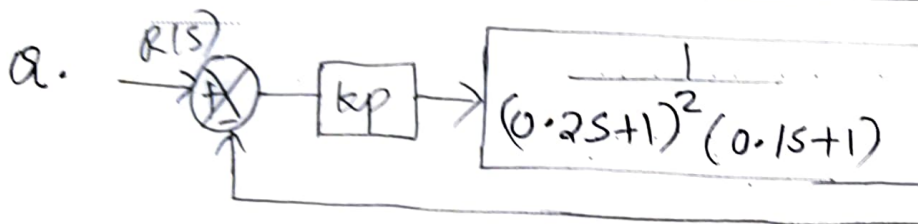
$$TF = \frac{G'}{1 + G'} = \frac{42.5s + 27}{1.572s^4 + 9.434s^3 + 51.8s^2 + 27s + 27}$$

PID control

$$k_p = 0.6 k_{cr} = 6$$

$$T_i = \frac{P_{cr}}{2} = \frac{1.895}{2} = 0.9475$$

$$T_d = \frac{P_{cr}}{8} = \frac{1.895}{8} = 0.2368$$



$$CE \Rightarrow 1 + \frac{kp}{(0.2s+1)^2(0.1s+1)} = 0$$

$$0.004s^3 + 0.08s^2 + 0.5s + 1 + kp = 0$$

s^3	0.004	0.5
s^2	0.08	$1 + kp$

$$s^1 \quad \frac{0.04 - 0.004(1+kp)}{0.08}$$

$$s^0 \quad 1 + kp$$

$$0.004(1+kp) > 0.04$$

$$kp > 9$$

$$K_{cr} = 9$$

$$0.08s^2 + (1+kp) = 0$$

$$0.08s^2 + 10 = 0$$

$$s = \pm$$

Q. TF $G(s) = \frac{1}{(s+1)(s+3)(s+5)}$ the
placed under PID control

Ans. CE $1 + k_p G(s) = 0$
 $(s+1)(s+3)(s+5) + k_p = 0$
 $s^3 + 9s^2 + 23s + 15 + k_p$
Routh array

$$\begin{array}{ccc} s^3 & 1 & 23 \end{array}$$

$$\begin{array}{ccc} s^2 & 9 & 15 + k_p \end{array}$$

$$\begin{array}{ccc} s^1 & 192 - k \end{array}$$

$$\begin{array}{ccc} s^0 & 15 + k \end{array}$$

$$15 + k > 0 \quad k > -15$$

$$192 - k > 0 \quad k < 192$$

$$k_{cr} = 192$$

ρ_{cr}

$$k_d = k_p T_d = \frac{k_p}{8\rho_{cr}} =$$

$$TF = k_p + \frac{k_p}{s} + s k_d$$

$$TF = 115.2 + \frac{117.2}{s} +$$
